Data File Location:

https://github.com/tiwvis/MLBD-Projects/blob/main/orange_problem/stock_prices.csv

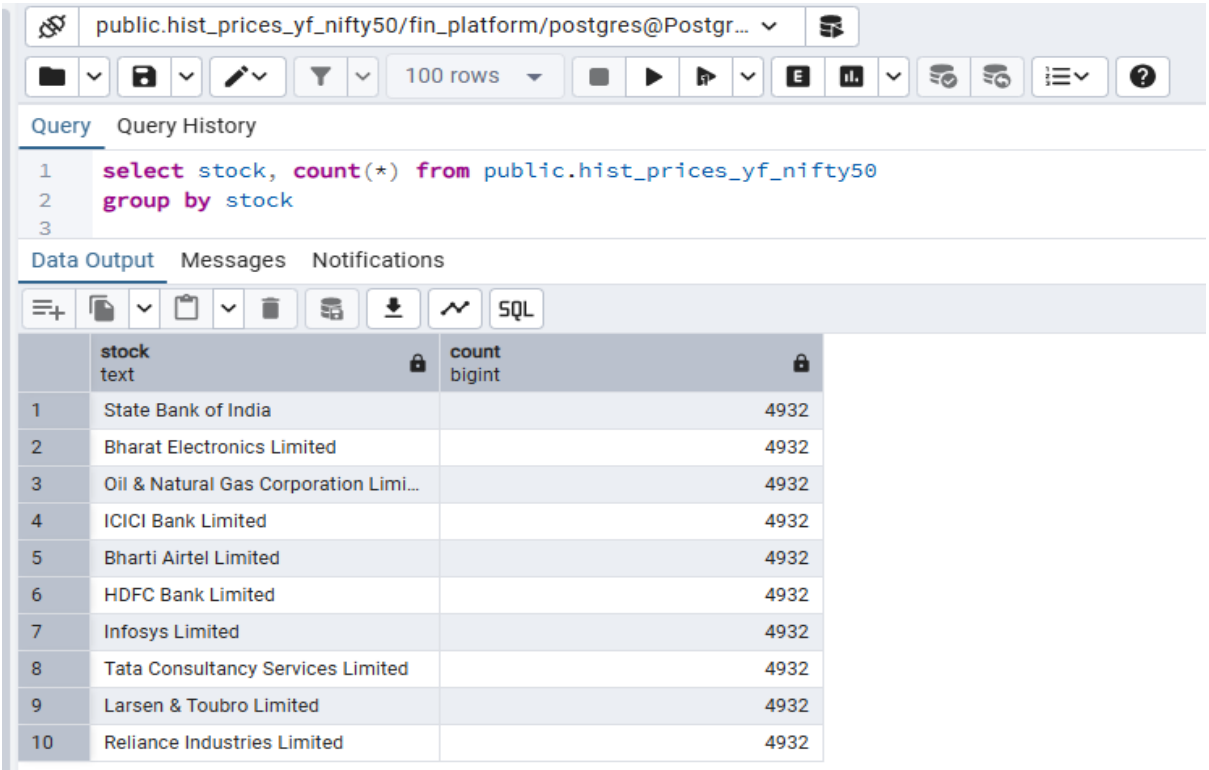
Raw data snapshot

symbol	stock	ticker	date	open	high	low	close	volume	dividend
HDFCBAN	HDFC Bank Limited	INE040A01034	09-10-2006	39.23592	39.65797	39.23592	39.44376	3071960	0
HDFCBAN	HDFC Bank Limited	INE040A01034	10-10-2006	39.448	40.08425	39.448	39.78733	2897020	0
HDFCBAN	HDFC Bank Limited	INE040A01034	11-10-2006	40.07153	41.22952	39.95913	40.96865	6854880	0
HDFCBAN	HDFC Bank Limited	INE040A01034	12-10-2006	41.02803	41.02803	40.2115	40.68445	9512260	0
HDFCBAN	HDFC Bank Limited	INE040A01034	13-10-2006	41.4713	42.41508	41.4416	41.89759	6900340	0
HDFCBAN	HDFC Bank Limited	INE040A01034	16-10-2006	42.37479	45.17221	42.07787	44.48717	19710100	0
HDFCBAN	HDFC Bank Limited	INE040A01034	17-10-2006	44.49565	45.04707	42.23057	42.74806	30601940	0
HDFCBAN	HDFC Bank Limited	INE040A01034	18-10-2006	43.64729	43.64729	41.38009	41.56249	27611500	0

Data description:

symbol (text) – NSE stock symbol for stock
stock (text) – Name of the stock
ticker (text)- NSE identifier of the stock
price_date (date): The specific trading day or timestamp for the record.
open (numeric): The price of the asset when the market opened.
high (numeric): The highest price reached during that trading session.
low (numeric): The lowest price reached during that trading session.
close (numeric): The final trading price when the market closed for the day.
volume (integer) : Total number of shares available for trading
dividends (numeric) – Dividend paid by the company

Data Size: *Contains 20 years of historical returns of stocks*



	stock text	count bigint
1	State Bank of India	4932
2	Bharat Electronics Limited	4932
3	Oil & Natural Gas Corporation Limi...	4932
4	ICICI Bank Limited	4932
5	Bharti Airtel Limited	4932
6	HDFC Bank Limited	4932
7	Infosys Limited	4932
8	Tata Consultancy Services Limited	4932
9	Larsen & Toubro Limited	4932
10	Reliance Industries Limited	4932

Feature Engineering Tasks:

Calculate following derived fields:

day, month, year, is_quarter_end, open-close, low-high, target

Sample code for calculation of derived fields:

```
splitted = df['price date'].str.split('/', expand=True)
df['day'] = splitted[1].astype('int')
df['month'] = splitted[0].astype('int')
df['year'] = splitted[2].astype('int')
df['is_quarter_end'] = np.where(df['month']%3==0,1,0)
df['open-close']  = df['open'] - df['close']
df['low-high']  = df['low'] - df['high']
df['target'] = np.where(df['close'].shift(-1) > df['close'], 1, 0)
```

Prediction data: Perform the prediction on the following data set:

Data Location:

https://github.com/tiwvis/MLBD-Projects/blob/main/orange_problem/prediction_data.csv

symbol	stock	ticker	date	open	high	low	close	volume	dividend
HDFCBAN	HDFC Bank Limited	INE040A0	14-10-2026	707	716.5	705.1	714	18838639	0
RELIANCE	Reliance Industries Limited	INE002A0	14-10-2026	1158.9	1168.3	1152.2	1166.7	10988483	0
LT	Larsen & Toubro Limited	INE018A0	14-10-2026	3627.5	3678	3592.9	3675	2254840	0
ICICIBANK	ICICI Bank Limited	INE090A0	14-10-2026	1339	1364.6	1329.4	1360.2	5765182	0
BHARTIAR	Bharti Airtel Limited	INE397D0	14-10-2026	1756.2	1806.5	1756.2	1806.5	5942868	0
SBIN	State Bank of India	INE062A0	14-10-2026	941	957	938.6	957	7120906	0
ONGC	Oil & Natural Gas Corporation Limited	INE213A0	14-10-2026	221.2	222.65	219.02	222.65	5893758	0
BEL	Bharat Electronics Limited	INE263A0	14-10-2026	373	380	369.4	380	8041157	0
INFY	Infosys Limited	INE009A0	14-10-2026	1029.2	1065	1027	1062.1	21781113	0
TCS	Tata Consultancy Services Limited	INE467B0	14-10-2026	2145.494	2257.3	2143.604	2257.3	11048193	0

Sample Output (Target =1 : Buy, Target = 0: Sell):

State Bank of India	**Buy**
Bharat Electronics Limited	**Sell**
Oil & Natural Gas Corporation Limited	**Sell**
ICICI Bank Limited	**Buy**
Bharti Airtel Limited	**Buy**
HDFC Bank Limited	**Sell**
Infosys Limited	**Buy**
Tata Consultancy Services Limited	**Sell**
Larsen & Toubro Limited	**Buy**
Reliance Industries Limited	**Sell**

Problem Statement:

- Load the stock data
- Perform feature engineering (as mentioned above)
- Perform EDA
- Train the model – XGBoost
    - Train the model for **all the 10 stocks** with following features:
    - Prediction variables – ***Stock, open-close, low_high, is_quarter_end***
    - Predicted variable - ***target***
- Predict the result (for data provided for prediction)
- Output the result in format specified

Submission:

1) Code
2) Analysis and Results (PDF file)
    - Snapshots of EDA
    - Result/Output